# Supplement 1: Searches for VWD diagnosis guidelines


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We will combine these search terms to construct search strategies using **operators like AND and OR**.

We will use OR to combine the terms within each concept (e.g. Patients with VWD, bleeding assessment tools, VWF activity assays identified with a light grey shade in Table 1), to make sure we capture all the relevant articles to that specific concept, regardless of how the researchers labeled it. For example, when searching for articles about desmopressin trials, the search string will be "Deamino Arginine Vasopressin or desmopressin or DDAVP".

We will use AND to combine different concepts, to make sure to capture articles that are addressing the questions of interest. For example, for the question regarding bleeding assessment tools, we will combine the concepts "patients with VWD AND intervention bleeding assessment tools".

For some questions, the search will have both operators. For example, for the question regarding Patients with suspected Type 2A, 2B or 2M in need of additional testing, the strategy will combine patients with VWD AND (intervention VWF multimer testing) OR (intervention Collagen binding testing). Table 2 shows details of this.

We will also use other strategies for some concepts, such as **proximity of words**. For example, for patients with VWD, we will also search for abstracts that have the terms "Von Willebrand" close to the terms "disease OR disorder".

In addition, we will use **truncations and wild cards**- which are illustrated in the table. A truncation searches for all terms that are the same to the one specified up to the truncation. For example, searching for "bleed*" and truncating it will search for bleed, bleeding, bleeder, etc.

Finally, we will use **study design filters** to only capture randomized clinical trials and any type of observational studies comparing at least 2 of the options of interest, in addition to a filter to only capture articles related to diagnosis.

We will not limit the search by outcomes measured in the studies, date, or language of publication.

Table 1: Search terms used for each of the concepts

| Term | It also includes |
|---|---|
| Patients with VWD | |
| Von Willebrand Disease | |
| Von Willebrand Disorder | |
| VWD | |
| Intervention: Bleeding assessment tools | |
| Bleeding assessment tools | |
| Bleeding score | |
| Screening tool | |
| questionnaire | |
| Interview | |
| Intervention: VWF activity assays | |
| Von Willebrand Factor | |
| VWF | |
| Ristocetin cofactor | VWF:RCo |
| VWF:G1b | VWF:G1bM<br>VWF:GP1bR |
| Assay | |
| Glycoprotein 1b | |
| Glycoprotein Ib | |
| Intervention: Diagnosis exclusion | |
| Phenotype | |
| Evolving | |
| Aging | |
| Intervention: VWF propeptide trial | |
| VWF Propeptide | |
| VWFpp | |
| Intervention: Desmopressin trial | |
| Desmopressin trial | |
| Desmopressin challenge | |
| DDVAP | |
| Deamino Arginine Vasopressin | |
| Intervention: VWF multimer testing | |
| multimer | |
| Intervention: Collagen binding testing | |
| Collagen binding | |
| VWF:CB | |
| Intervention: Ristocetin induced platelet binding(RIPA) testing | |
| Ristocetin induced platelet binding | |
| RIPA | |
| PT-VWD | |
| Intervention: Genetic testing | |
| Genetic test | Genetic Assay<br>Genetic Screen<br>Genetic Assessment |
| Mutation | |
| Variant | |
| Genotype | |
| Gene | |

| | |
|---|---|
| Inheritance | |
| Intervention: VWF:FVIII binding | |
| VWF:FVIII binding | |
| VWF:FVIIIB binding | |
| Factor 8 binding | |

Table 2: Concepts used for each question

| **Question** | **Concepts** |
|---|---|
| BAT in patients suspected of VWD | patients with VWD AND Intervention Bleeding assessment tools |
| Patients suspected of VWD with negative BAT | patients with VWD AND intervention Bleeding assessment tools |
| Essays in patients with suspected VWD | patients with VWD AND intervention VWF activity assays |
| Patients with a historic diagnosis of VWD but who now have normal VWF levels | patients with VWD AND intervention Diagnosis exclusion |
| Patients with abnormal initial screen of VWF:Ag, VWF:RCo, and suspected Type 1 VWD cut-off values | patients with VWD AND intervention VWF activity assays |
| Propeptide levels and desmopressin trials in patients suspected of Type 1 VWD | patients with VWD AND (intervention VWF propeptide trial) OR (intervention Desmopressin trial) |
| Patients with abnormal initial screen of VWF:Ag, VWF:RCo and a low RCo:Ag ratio | patients with VWD AND intervention VWF activity assays |
| Patients with suspected Type 2A, 2B or 2M in need of additional testing | patients with VWD AND (intervention VWF multimer testing) OR (intervention Collagen binding testing) |
| Patients suspected to have Type 2B VWD | patients with VWD AND (intervention Ristocetin induced platelet binding) OR (intervention Genetic testing) |
| Patients suspected to have Type 2N in need of additional testing | patients with VWD AND (intervention VWF:FVIII binding) OR (intervention Genetic testing) |

Example of search strategy: Bleeding assessment tools in patients suspected of VWD.

1. exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.
2. (bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp.
3. (bleed* adj4 (assess* or screen* or tool*)).mp.a
4. 2 or 3
5. 1 and 4

*The search strategy contains the following, which were not described above: “exp term /”, searches for all the terms indexed under the term (for example, exploding the term “Von Willebrand Diseases” searches also for VWD type 1, type 2, type 3 etc; “mp.”, searches for the term in multiples fields, including title, abstract, keywords, etc; “adj4”, is the tool to search for terms close to one another (in line 1, we searched for von willebrand term within 4 words of any of the 2 second terms (disease or disorder).

**PICO 1: In Patients with suspected VWD - Using a bleeding assessment tool (any tool) to screen for VWD VS Non-standardized clinical assessment (not using a bleeding assessment tool)**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | (bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp. | 784636 |
| 3 | (bleed* adj4 (assess* or screen* or tool*)).mp. | 2740 |
| 4 | 2 or 3 | 787072 |
| 5 | 1 and 4 | 314 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | (bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp. | 1094952 |
| 3 | (bleed* adj4 (assess* or screen* or tool*)).mp. | 5047 |
| 4 | 2 or 3 | 1099275 |
| 5 | 1 and 4 | 853 |

**PICO 2: Patients suspected of VWD with negative BAT (men and children) - Blood testing VS No blood testing**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | (bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp. | 784636 |
| 3 | (bleed* adj4 (assess* or screen* or tool*)).mp. | 2740 |
| 4 | 2 or 3 | 787072 |
| 5 | (blood adj4 test*).mp. | 67402 |
| 6 | 4 or 5 | 850438 |
| 7 | 1 and 6 | 913 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | (bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp. | 1094952 |
| 3 | (bleed* adj4 (assess* or screen* or tool*)).mp. | 5047 |
| 4 | 2 or 3 | 1099275 |
| 5 | (blood adj4 test*).mp. | 113728 |
| 6 | 4 or 5 | 1205027 |
| 7 | 1 and 6 | 1258 |

**PICO 3: In Patients with suspected VWD - VWF:RCo (Automated and non-automated assays) VS Newer assays that reflect the platelet binding activity of VWF function (ie: VWF:G1bM, VWF:GP1bR).**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | exp von willebrand factor/ or VWF.mp. or ristocetin cofactor.mp. or GPIb*.mp. or glycoprotein 1b.mp. or glycoprotein Ib.mp. or assay*.mp. or activity.mp. | 3448167 |
| 3 | 1 and 2 | 3376 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | exp von willebrand factor/ or (VWF or ristocetin cofactor or glycoprotein 1b or glycoprotein Ib or GPIb* or assay*).mp. | 1602981 |
| 3 | 1 and 2 | 5758 |

**PICO 4: Patients with a historic diagnosis of VWD but who now have normal VWF levels - Remove the diagnosis of VWD VS Reconsider another diagnosis**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | (Phenotype or evolving or aging).mp. | 804294 |
| 3 | 1 and 2 | 461 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | (Phenotype or evolving or aging).mp. | 1189124 |
| 3 | 1 and 2 | 1034 |

Notes: Subgroup: Bleeding phenotype

**PICO 5: Patients with abnormal initial screen of VWF:Ag, VWF:RCo, and suspected Type 1 VWD - Diagnosis of Type 1 VWD at VWF:Ag and/or VWF:RCo < 0.30 IU/mL VS Diagnosis of Type 1 VWD at VWF :Ag and/or VWF:RCo < the reference range of your lab which is usually 0.50 IU/mL**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | exp von willebrand factor/ or VWF.mp. or ristocetin cofactor.mp. or GPIb*.mp. or glycoprotein 1b.mp. or glycoprotein Ib.mp. or assay*.mp. or activity.mp. | 3448167 |
| 3 | 1 and 2 | 3376 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | exp von willebrand factor/ or (VWF or ristocetin cofactor or glycoprotein 1b or glycoprotein Ib or GPIb* or assay*).mp. | 1602981 |
| 3 | 1 and 2 | 5758 |

Notes: VWF:RCo means activity essays in general ( ie: VWF:G1bM, VWF:GP1bR)

**PICO 6: Patients suspected of Type 1 VWD – VWFpp VS Desmopressin trial with 1 and 4 hour bloodwork**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | (propeptide or VWFpp).mp. | 5897 |
| 3 | Deamino Arginine Vasopressin/ or desmopressin.mp. or DDAVP.mp. | 5690 |
| 4 | (challenge or trial).mp. | 1383647 |
| 5 | 3 and 4 | 736 |
| 6 | 2 or 5 | 6625 |
| 7 | 1 and 6 | 182 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | (propeptide or VWFpp).mp. | 7864 |
| 3 | "argipressin[1 deamino]"/ or desmopressin.mp. or DDAVP.mp. | 13200 |
| 4 | (challenge or trial).mp. | 2282560 |
| 5 | 3 and 4 | 2119 |
| 6 | 2 or 5 | 9976 |
| 7 | 1 and 6 | 535 |

**PICO 7: Patients with abnormal initial screen of VWF:Ag, VWF:RCo and a low RCo:Ag ratio - RCo:Ag <0.5 VS Other cut-offs (ie: 0.6 or 0.7)**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | exp von willebrand factor/ or VWF.mp. or ristocetin cofactor.mp. or GPIb*.mp. or glycoprotein 1b.mp. or glycoprotein Ib.mp. or assay*.mp. or activity.mp. | 3448167 |
| 3 | 1 and 2 | 3376 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | exp von willebrand factor/ or (VWF or ristocetin cofactor or glycoprotein 1b or glycoprotein Ib or GPIb* or assay*).mp. | 1602981 |
| 3 | 1 and 2 | 5758 |

Notes: VWF:RCo means activity essays in general ( ie: VWF:G1bM, VWF:GP1bR)

**PICO 8: Patients with suspected Type 2A, 2B or 2M in need of additional testing - VWF multimer analysis VS VWF:CB**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | (multimer or Collagen binding or VWF:CB).mp. | 3926 |
| 3 | 1 and 2 | 477 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | (multimer or Collagen binding or VWF:CB).mp. | 5457 |
| 3 | 1 and 2 | 941 |

**PICO 9: Patients suspected to have Type 2B VWD - Ristocetin induced platelet binding(RIPA) VS Genetic testing (mutation analysis)**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | (ripa or ristocetin induced platelet binding or PT-VWD).mp. | 501 |
| 3 | ((genetic or mutation or variant or genotype or gene or inheritance) adj4 (test* or assay* or screen* or assess*)).mp. | 163485 |
| 4 | 2 or 3 | 163976 |
| 5 | 1 and 4 | 275 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | (ripa or ristocetin induced platelet binding or PT-VWD).mp. | 873 |
| 3 | ((genetic or mutation or variant or genotype or gene or inheritance) adj4 (test* or assay* or screen* or assess*)).mp. | 206516 |
| 4 | 2 or 3 | 207360 |
| 5 | 1 and 4 | 517 |

**PICO 10: Patients suspected to have Type 2N in need of additional testing - VWF:FVIII binding VS Genetic testing.**

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 6470 |
| 2 | ((factor VIII or VWF:FVIII or factor 8) adj4 binding).mp. | 323 |
| 3 | ((genetic or mutation or variant or genotype or gene or inheritance) adj4 (test* or assay* or screen* or assess*)).mp. | 163485 |
| 4 | 2 or 3 | 163802 |
| 5 | 1 and 4 | 253 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp. | 10275 |
| 2 | ((factor VIII or VWF:FVIII or factor 8) adj4 binding).mp. | 426 |
| 3 | ((genetic or mutation or variant or genotype or gene or inheritance) adj4 (test* or assay* or screen* or assess*)).mp. | 206516 |
| 4 | 2 or 3 | 206925 |
| 5 | 1 and 4 | 436 |

## Study Design Filters

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 29 | limit 28 to human | 3276 |
| 30 | (randomized controlled trial or controlled clinical trial).pt. or (randomized or placebo).ab. or drug therapy.fs. or (randomly or trial or groups).ab. | 4337050 |
| 31 | epidemiologic studies/ or exp case control studies/ or exp cohort studies/ or Cross-sectional studies/ | 2225583 |
| 32 | case control.tw. | 111970 |
| 33 | (cohort adj (study or studies)).tw. | 166398 |
| 34 | cohort analy$.tw. | 6624 |
| 35 | (Follow up adj (study or studies)).tw. | 46061 |
| 36 | (Observational adj (study or studies)).tw. | 87158 |
| 37 | (longitudinal or retrospective or cross sectional).tw. | 921384 |
| 38 | or/31-36 | 2366260 |
| 39 | (registr* or database).mp. | 468967 |
| 40 | (sensitiv: or diagnos:).mp. or di.fs. | 5730164 |
| 41 | 30 or 38 or 39 or 40 | 10134762 |
| 42 | 29 and 41 | 1939 |

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

| # | Searches | Results |
|---|---|---|
| 29 | limit 28 to human | 5475 |
| 30 | di.fs. or predict:.tw. or specificity.tw. | 4873662 |
| 31 | random*.ab. or random*.ti. or (clinical adj trial*).hw,ab,ti. or exp 'health care quality'/ | 4680717 |
| 32 | exp disease course/ or risk:.mp. or diagnos:.mp. or follow-up.mp. or ep.fs. or outcome.tw. | 11189515 |
| 33 | 30 or 31 or 32 | 14073954 |
| 34 | 29 and 33 | 3781 |

Supplement 2: Excluded studies

Patients with normalized VWF levels:


1. A. G. Casonato L, Cattini, M. G., Sartorello, F., Pontara, E., Padrini, R., Bertomoro, A., Daidone, V., Pagnan, A. Type 1 von Willebrand disease due to reduced von Willebrand factor synthesis and/or survival: observations from a case series. 2010.
2. Abbott D, Paola, J. D. VWD type 1: A calculated diagnosis. *Blood.* 2008;111(8):3919-3920.
3. Abshire TC. Diagnosis of type 1 VWD: Can the clinical history trump laboratory findings? *Pediatric Blood and Cancer.* 2009;52(1):7-8.
4. Batty P, Hodgetts, J. N., Brown, C., Uprichard, J. Impact of changes in laboratory diagnostic criteria in the classification of von willebrand disease. *Research and Practice in Thrombosis and Haemostasis.* 2017;1 (Supplement 1):713.
5. C. D. R. Seaman MV. The Association of Aging With Von Willebrand Factor Levels and Bleeding Risk in Type 1 Von Willebrand Disease. 2018.
6. Castaman G, Rodeghiero, F. Platelet ristocetin cofactor (RiCof) in patients with type I "platelet normal" von Willebrand disease correlates with plasma RiCof peak after desmopressin infusion. *Thrombosis Research.* 1995;77(4):383-384.
7. Chen SF, Xia, Z. L., Han, J. J., Wang, Y. T., Wang, J. Y., Pan, S. D., Wu, Y. P., Zhang, B., Li, G. Y., Du, J. W., Gao, H. Q., De Groot, P. G., De Laat, B., Hollestelle, M. J. Increased active von Willebrand factor during disease development in the aging diabetic patient population. *Age.* 2013;35(1):171-177.
8. Christopherson PA, Flood, V. H., Friedman, K. D., Gill, J. C., Haberichter, S. L., Montgomery, R. R., Abshire, T. C. Impact of age and gender on bleeding score in type 1 VWD using the ISTH-BAT. *Blood Conference: 56th Annual Meeting of the American Society of Hematology, ASH.* 2014;124(21).
9. E. F. C. Grabowski MA, Van Cott, E. M. Assessment of a Cohort of Primarily Pediatric Patients with a Presumptive Diagnosis of Type 1 von Willebrand Disease with a Novel High Shear Rate, Non-citrated Blood Flow Device.
10. Favaloro EJ. A duplex issue: (i) time to re-appraise the diagnosis and classification of von Willebrand disorder, and (ii) clarification of the roles of von Willebrand factor collagen binding and ristocetin cofactor activity assays. *Haemophilia : the official journal of the World Federation of Hemophilia.* 2002;8(6):828-831.
11. Favaloro EJ, Bonar, R., Marsden, K. Lower limit of assay sensitivity: an under-recognised and significant problem in von Willebrand disease identification and classification. *Clinical laboratory science : journal of the American Society for Medical Technology.* 2008;21(3):178-183.
12. Fleming AF. Ethnic variation in von Willebrand factor levels can influence the diagnosis of von Willebrand disease. *Clinical and Laboratory Haematology.* 2003;25(6):413.
13. Jaafar S, Yee, T. T. A lookback at VWD type 2A and 2M classification in a large comprehensive haemophilia centre. *Haematologica.* 2017;102 (Supplement 2):96.
14. Konkle BA. Von Willebrand factor and aging. *Seminars in Thrombosis and Hemostasis.* 2014;40(6 Supplement 01):640-644.
15. Laffan M. Can you grow out of von Willebrand disease? 2017.
16. M. G. Borghi G, Mezzasoma, A. M., Falcinelli, E., Gresele, P. Variations of hemostatic parameters in relation to age in type 1 von willebrand disease. 2016.
17. Mannucci PM, Lombardi, R., Bader, R. Heterogeneity of type I von Willebrand disease: Evidence for a subgroup with an abnormal von Willebrand factor. *Blood.* 1985;66(4):796-802.
18. Miesbach W, Berntorp, E. When von Willebrand disease comes into age - a matter of change? *Eur J Haematol.* 2011;86(6):496-501.

19. Morita E, Soundar, E., Bruzdoski, K., Kostousov, V., Teruya, J., Hui, S. K. Can a one-time von willebrand testing rule out type 1 von willebrand disease in young women with menorrhagia? *Research and Practice in Thrombosis and Haemostasis.* 2017;1 (Supplement 1):718.
20. Onundarson PT, Gudmundsdottir, B. R., Marder, V. J. On the significance of marginally low von Willebrand factor. *Thrombosis and Haemostasis.* 2008;100(6):1213-1214.
21. Quiroga T, Goycoolea, M., Zuniga, P., Belmont, S., Panes, O., Pereira, J., Mezzano, D. NHLBI guidelines and the percentile 2.5th of plasma VWF for the diagnosis of type 1 VWD. Comparison in 4517 referred patients and 280 selected patients with conclusive bleeding history. *Journal of Thrombosis and Haemostasis.* 2013;2):414.
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23. Sampson M, Cheng, D., Fargo, J. The effect of age at diagnosis of type i von willebrand disease on diagnostic lab values: A pediatric perspective. *Pediatric Blood and Cancer.* 2017;64 (Supplement 1):S10.
24. Sanders YV, Giezenaar, M. A., Laros-Vangorkom, B., Meijer, K., Van Derbom, J. G., Cnossen, M. H., De Meris, J., Nijziel, M. R., Ypma, P. F., Fijnvandraat, C. J., Eikenboom, J., Mauser-Bunschoten, E. P., Leebeek, F. W. Aging in von Willebrand disease: A nationwide multicenter cross-sectional study with a retrospective longitudinal substudy. *Haemophilia.* 2014;2):21-22.
25. Sanders YV, Laros-Van Gorkom, B. A. P., Meijer, K., Van Der Bom, J. G., Cnossen, M. H., Nijziel, M. R., Fijnvandraat, K., Eikenboom, J., Mauser-Bunschoten, E. P., Leebeek, F. W. G. Von Willebrand disease and aging: An evolving phenotype. *Haemophilia.* 2014;3):110.
26. Siboni SM, Biguzzi, E., Mistretta, C., Caiani, V., Mannucci, P. M., Bucciarelli, P., Peyvandi, F. Relationship between age at first bleeding and baseline Factor VIII plasma levels in patients with severe forms of von Willebrand disease (VWD). *Thrombosis Research.* 2014;2):S23.

VWF activity cut-off for VWD type 1 diagnosis:

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Supplement 3: Risk of bias

| Author, year | Study Participation | Study Attrition | Prognostic Factor Measurement [a] | Outcome Measurement [a] | Study Confounding [b] | Statistical Analysis and Reporting |
|---|---|---|---|---|---|---|
| Sanders, 2014 | Low | Low | High | Low | Low | Low |
| Borghi, 2017 | Low | Low | High | Low | High | Low |
| Nummi, 2017 | Low | Low | High | Low | High | Low |
| Lavin, 2017 | Low | Low | High | Low | High | Low |
| Rydz, 2015 | Moderate | Low | High | Low | High | Low |
| Abou-Ismail, 2017 | Low | Low | High | Low | High | Low |

a. Bleeding symptoms not measured in patients with normalized levels

b. Study confounding high in studies that did not adjust for comorbidities while measuring the outcome of interest

| Author, year | Patient selection Risk of bias | Index test Risk of bias | Reference standard Risk of bias | Flow and timing Risk of bias |
|---|---|---|---|---|
| Lavin, 2017 | Low | Moderate | Low | Low |
| Flood, 2016 | Low | Moderate | Low | Low |
| Bucciarelli, 2015 | Low | Moderate | Low | Low |
| Quiroga, 2014 | Low | Low | Moderate | Low |
| Bowman, 2009 | Low | Low | Low | Low |
| Tosetto, 2007 | Low | Moderate | Low | Low |
| James, 2007 | Low | Moderate | Low | Low |
| Goodeve, 2007 | Low | Moderate | Low | Low |
| Eikenboom, 2006 | Low | Moderate | Low | Low |

| Author, year | Patient selection risk of bias | Index test risk of bias | Reference standard risk of bias | Flow and timing risk of bias |
|---|---|---|---|---|
| Vangenechten, 2018 | High | Low | Low | Low |
| de Maistre, 2014 | High | Low | Low | Low |
| Chen, 2011 | Low | Low | Low | Low |
| James, 2007 | High | Low | Low | Low |
| Caron, 2006 | High | Low | Low | Low |
| Adcock, 2006 | Low | Low | Low | Low |